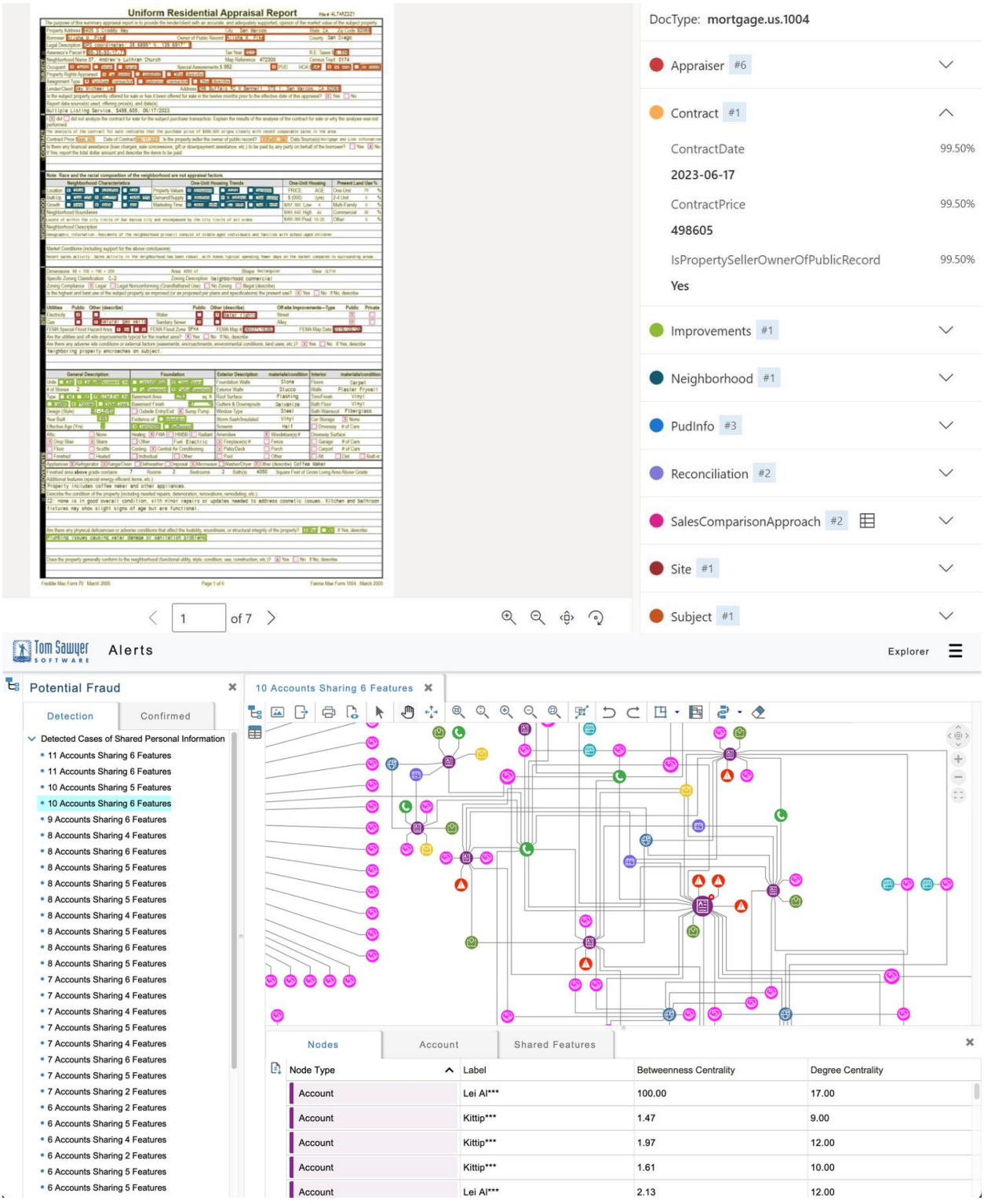
Project concept: LandShield BD — Intelligent Land Fraud Detection & Evidence Intelligence System

If your goal is to stand out at LandTech 2026, I would build a project where fraud detection is the core engine, supported by several connected features that make it more than a basic land-record management system.

The strongest approach is to detect suspicious relationships across land documents, ownership histories, mutations, plot boundaries and transaction timelines—then explain why a case looks suspicious.

One important distinction: you can create a novel combination of existing technologies, but we cannot honestly claim that a feature has never existed anywhere without extensive prior-art research. The opportunity is to make the combination unusually strong for the Bangladesh land context.





1. The core idea

LandShield BD

Explainable, multi-signal land fraud detection

Input: scanned khatian/porcha, deeds, mutation records, plot details and transaction history.

Output: a fraud-risk assessment, connected evidence, a timeline of inconsistencies, and recommended verification steps—not an automatic legal verdict.

The key differentiator is that your system should not merely ask, “Does this document look fake?” It should ask:

- Does this seller's claimed ownership match the available ownership chain?
- Has the same parcel apparently been sold to multiple buyers?
- Do plot identifiers, areas or ownership shares conflict across documents?
- Does a mutation appear inconsistent with the previous recorded owner?
- Are several suspicious transactions connected through the same people, documents or parcels?
- Which pieces of evidence support each alert, and what additional verification is needed?

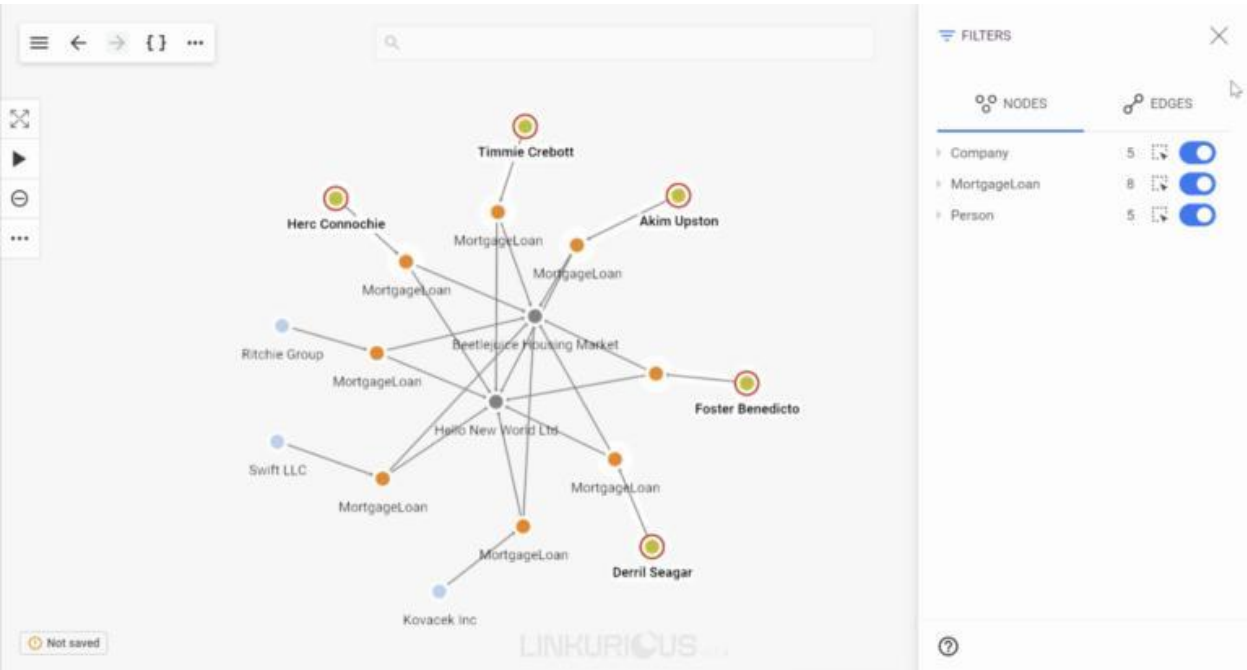
2. Strong features that work together

A quick scan of existing work shows that land-record fraud detection, OCR, ownership-chain checking and blockchain-based verification have already been explored individually or in combinations. So adding a QR code, AI chatbot or blockchain alone will not make your project novel.



A-42-018/LandSheild-BD · GitHub
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Instead, build around these deeper capabilities.



1. Temporal ownership-graph investigation

Core feature

Model every parcel, owner, deed, mutation and transaction as a connected graph. Track how ownership changes over time and flag impossible or unsupported transitions.

Example: A seller transfers the same parcel to Buyer A and Buyer B, or sells more land than their recorded share allows.

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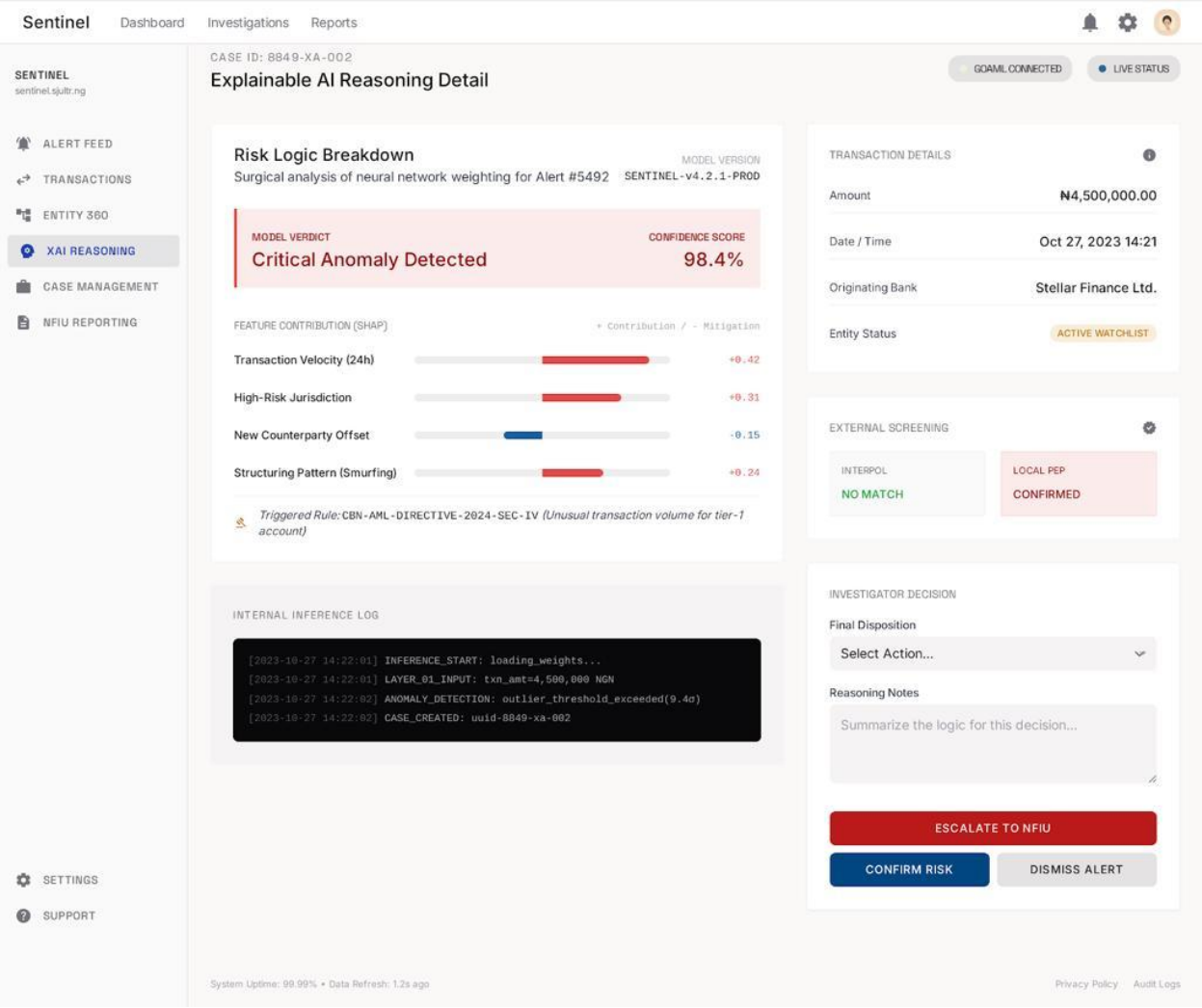
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2. Document contradiction engine

Extract Bangla text and structured fields from deeds, khatian/porcha and mutation papers. Compare the documents against one another, including names, dag numbers, survey types, area, ownership shares and dates.

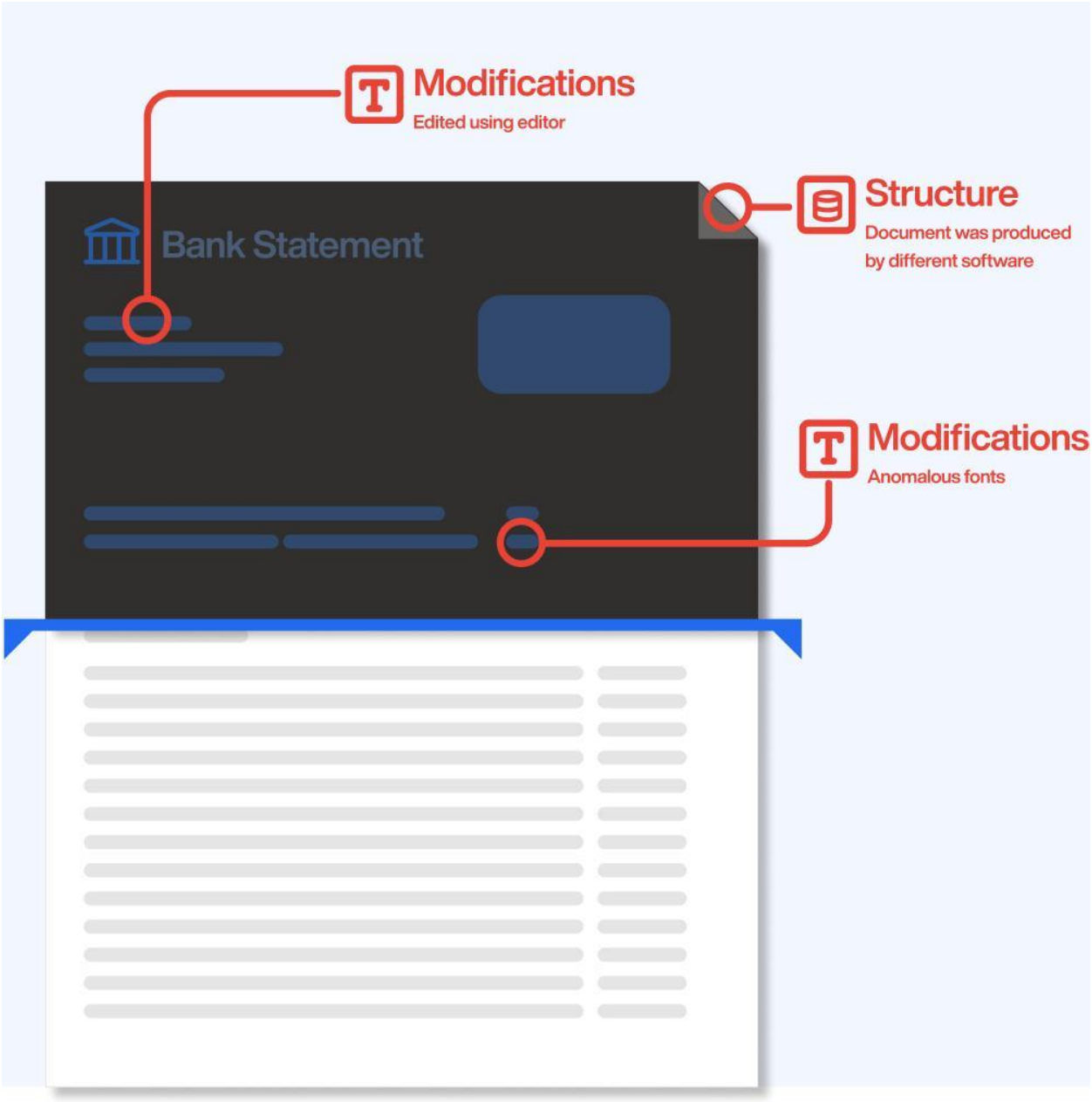
Standout detail: Show the exact conflicting fields side by side, with page and text-region references, instead of simply returning “document suspicious.”



3. Explainable fraud-risk engine

Combine hard rules with anomaly detection. Every alert must include the rule triggered, supporting records, severity and uncertainty.

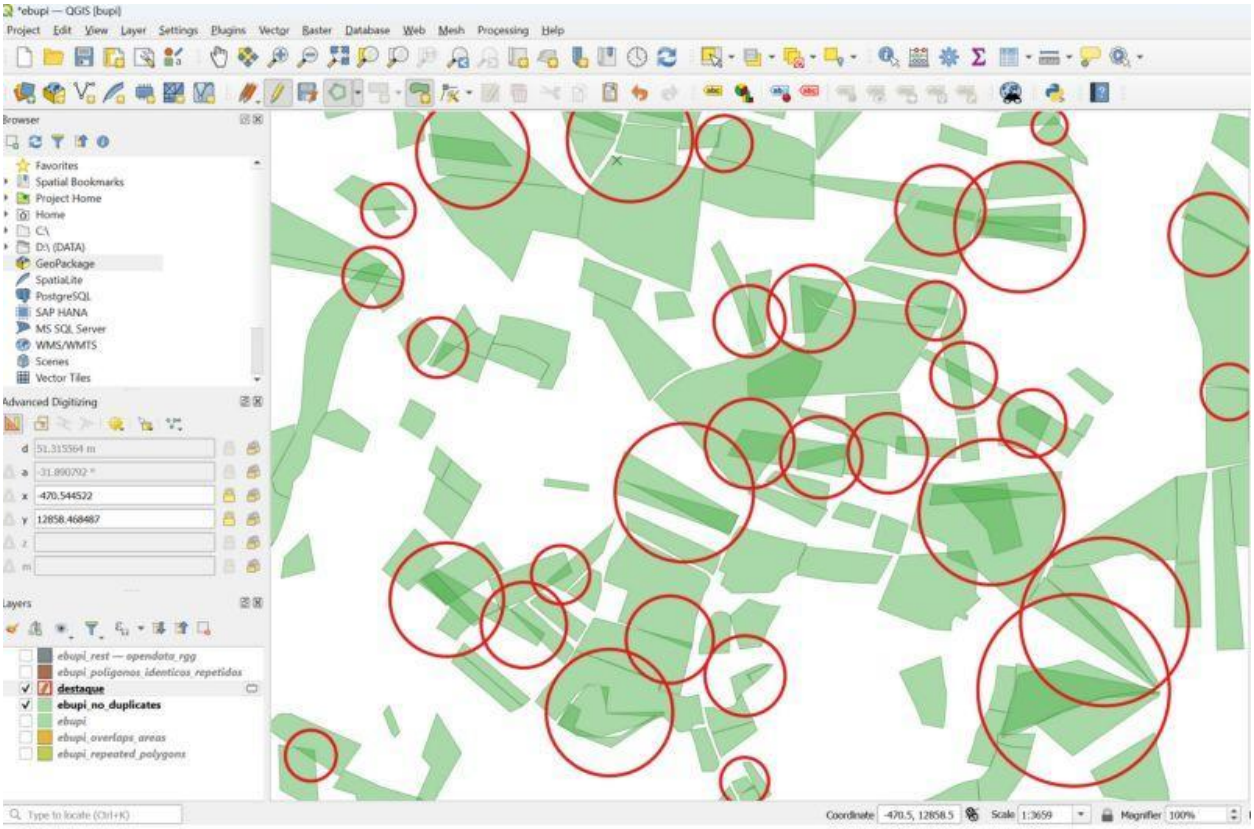
Example: “High risk: this parcel appears in two sale deeds dated 12 days apart, and the seller's recorded share does not cover the second transfer.”



4. Document alteration and provenance analysis

Compare document versions, cryptographic hashes and signed audit events. Where images permit, identify suspicious edits or inconsistent visual regions for human review.

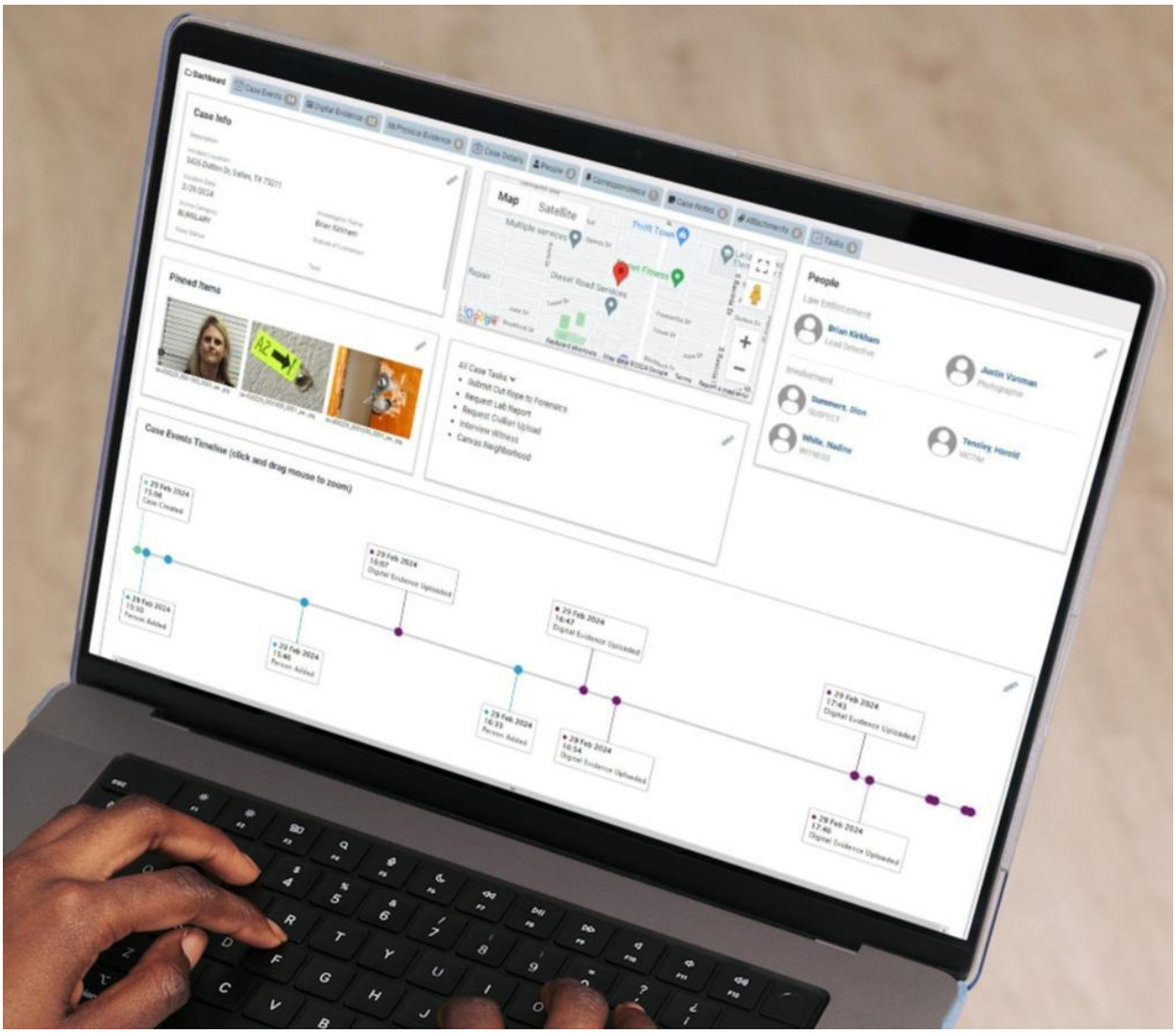
Important: A hash can show that a file changed after a trusted hash was recorded; it cannot prove that the original document was genuine. Image anomalies alone cannot establish forgery.



5. Parcel geometry and area-conflict detector

Link records to synthetic GIS parcels. Compare stated areas, plot identifiers and boundaries where reliable map data exists.

Example: Two records claim incompatible ownership over the same mapped parcel, or the claimed area exceeds the parcel's recorded area.



6. Adversarial fraud simulator and investigation copilot

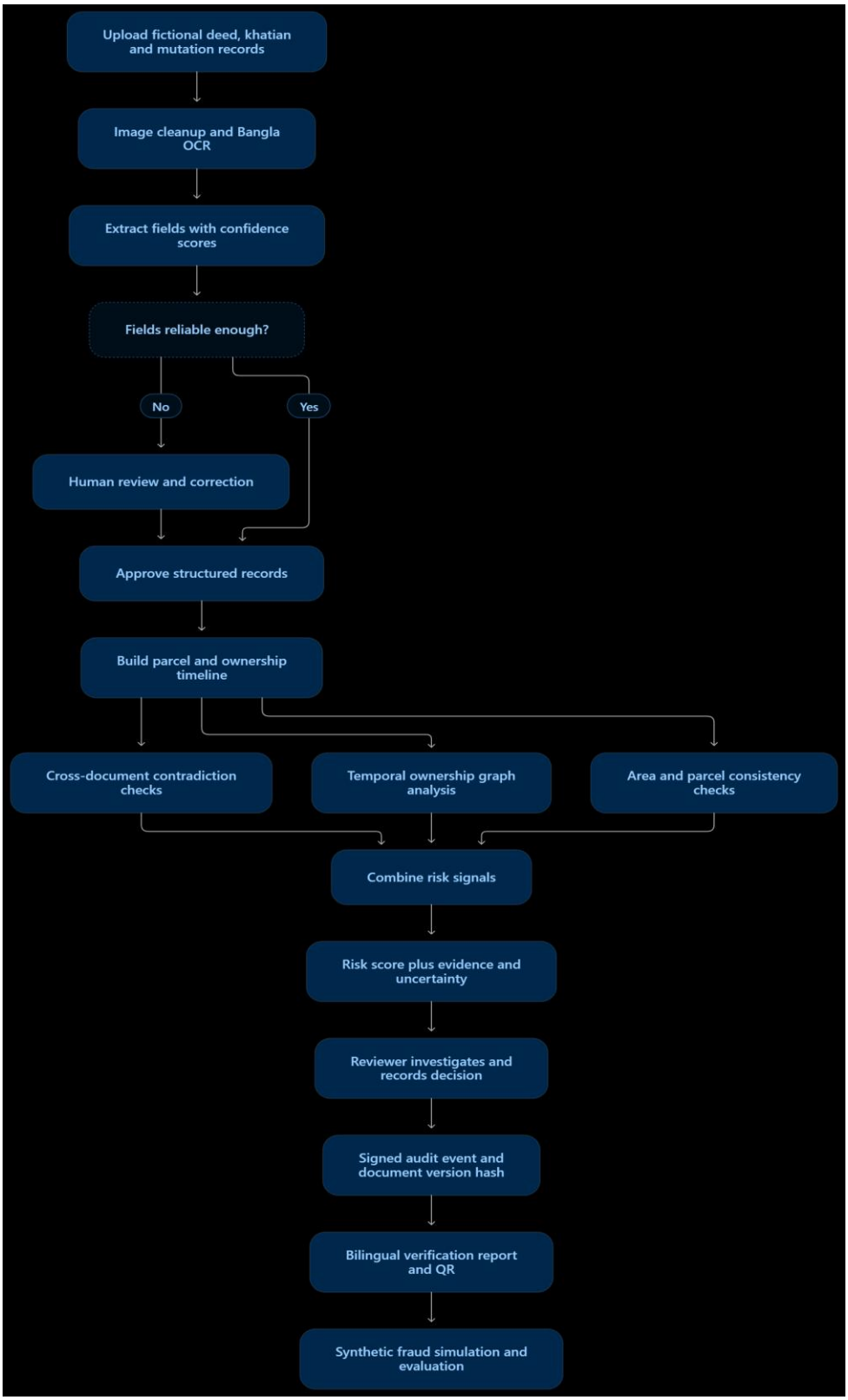
Generate fictional fraud scenarios—double sale, altered area, missing ownership link, inconsistent mutation date—and test whether your detector catches them. Then generate an evidence-based checklist in Bangla or English for a human reviewer.

Why it stands out: You can demonstrate measurable detection performance, show missed cases and false alarms, and prove that the system is more than a polished interface.

The last feature is my strongest candidate for your differentiator: a fraud detector that can test itself against generated attack scenarios and explain both what it caught and what it missed. Treat that as a proposed design direction, not a claim that no one has ever built it.

3. Full project workflow

Here is the end-to-end workflow I would present to the judges.



What happens in the fraud engine?

Use several independent detectors, then combine their results:

Detector	What it catches
Double-sale detector	Conflicting transfers involving the same parcel or share
Ownership-chain validator	A seller's claim has no supported link to the prior owner
Share and area validator	The claimed transfer exceeds the available ownership share or area
Cross-document comparison	Conflicting names, dates, plot numbers or survey details
Timeline anomaly detector	Suspicious transaction sequences or impossible dates
Graph-based link analysis	Separate suspicious transactions connected by shared parcels, documents or entities
Document integrity check	Changes to files after trusted recording, or suspicious image regions

Do not rely on a single AI model to make the final decision. Deterministic checks should catch explicit contradictions; AI can help extract difficult text and rank ambiguous patterns. The combined output should state which evidence is available and which information is missing.

4. The most ambitious feature: a counterfactual fraud investigation

This is the feature I would emphasize in your presentation because it turns the system from a simple risk dashboard into an investigation tool.

Imagine the system flags a parcel as high risk. Rather than only showing a score, it answers three questions:

1. Why was it flagged? Two deeds claim overlapping transfers, and the second seller's ownership share is insufficient.
2. What evidence would resolve the concern? A verified prior deed, an authoritative mutation record, or a human-confirmed correction may resolve specific inconsistencies.
3. What if the suspicious record is corrected? Recalculate the risk score and show which alerts disappear and which remain.

This is a counterfactual investigation: the reviewer can explore how changing a disputed fact affects the outcome. It must never silently change a real ownership record or treat a hypothetical correction as official.

5. Architecture and technology stack

Frontend

Next.js or React, Tailwind CSS, React Flow, Leaflet

Upload interface, ownership graph, parcel map, evidence timeline and reviewer dashboard.

AI and document processing

Python, FastAPI, OpenCV, Bangla OCR, structured extraction

Scan cleanup, text extraction, field confidence and document comparison.

Fraud analysis engine

Python, NetworkX, custom rules, scikit-learn if justified

Ownership-chain validation, graph analysis, anomaly scoring and counterfactual evaluation.

Data, security and audit

PostgreSQL, SHA-256, digital signatures, append-only audit events

Structured records, version integrity, reviewer actions and reproducible investigation history.

You do not need a blockchain to make this credible. A properly designed signed audit trail and trusted hash records are enough for a convincing prototype; explain the trust assumptions clearly.

6. How to build it for the hackathon

Do not attempt every feature at full scale. Build the following in order.

1. Phase 1 — Synthetic land-record dataset. Create linked fictional owners, parcels, deeds and mutation records, including clean cases and known fraud scenarios.
2. Phase 2 — Core fraud engine. Implement double-sale detection, ownership-chain checks, share/area validation and cross-document contradictions. This is the essential working feature.
3. Phase 3 — Visual investigation. Show the parcel's ownership timeline, related entities, flagged documents and evidence behind each alert.
4. Phase 4 — Document intelligence. Add Bangla OCR, field extraction, confidence values and manual correction before records enter the fraud engine.
5. Phase 5 — Novelty feature. Implement the adversarial scenario simulator and counterfactual risk analysis. Measure detection rate, false-positive rate and performance on held-out scenarios.
6. Phase 6 — Verification and reporting. Add version hashes, signed audit events, a privacy-preserving QR verification page and a Bangla/English report.

For the demo, prepare three cases:

- Clean case: consistent documents and a supported ownership chain.
- Double-sale case: two conflicting transfers involving the same parcel.
- Complex case: an ownership-share mismatch combined with inconsistent mutation information and a suspicious document version.

Show the judges how each case moves through the pipeline, which evidence triggered the result, and how the reviewer investigates it.

7. How to position its novelty

Your strongest defensible claim is not “the world's first AI land fraud detector.” Existing research already discusses blockchain-based land management, and other prototypes already demonstrate deed checking and ownership-chain analysis.


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Instead, position the proposed contribution as:

An explainable, self-evaluating land-fraud investigation system for Bangladesh that combines temporal ownership graphs, cross-document contradiction analysis, counterfactual risk testing and adversarial fraud simulation in one workflow.

That is a much more specific proposition. To substantiate novelty, compare it against existing projects and research, then document exactly what your implementation adds.

Also follow the LandTech rules: use dummy or organizer-approved sample data, do not scrape government sites or test live systems, and do not use real people's land records or personal identifiers without the required approval.

My final recommendation: Make fraud detection and ownership-graph investigation the main project, then prioritize counterfactual analysis and adversarial testing as your signature features. OCR, GIS, QR and cryptographic integrity should support that core—not distract from it.